

COURSEWORK 2 - BRIDGE ACCELERATION ANALYSIS

ZJUT 2026 International Summer Course - Vehicle-Centric Sensing (V2X)

Name: AL-DHANEBI ABDULMAJEED TAWFIK AHMED YAHYA Student ID: 312024705005

1. Selected Measurement Locations and Analysis Method

The vertical bridge acceleration response was analysed at $x = 7 \text{ m}$ and $x = 9 \text{ m}$. These two locations are close to the central region of the 15 m bridge and provide a useful comparison of the spatial variation of vibration response on either side of the midspan. The benchmark vehicle travels at 5 m/s , enters the bridge at $t = 3.0 \text{ s}$, and leaves at $t = 6.0 \text{ s}$. The sampling interval is 0.001 s , corresponding to a sampling frequency of 1000 Hz . Peak absolute acceleration and RMS acceleration were calculated from the 3-6 s bridge-crossing segment. For frequency-domain analysis, the signal mean was removed and a Hann window was applied before the FFT.

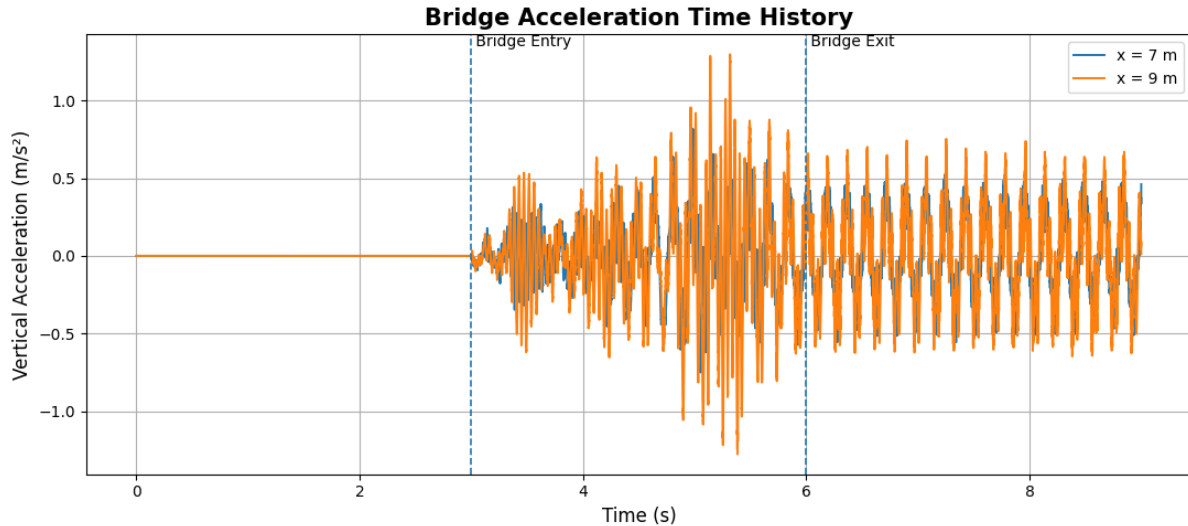


Figure 1. Full acceleration time history at $x = 7 \text{ m}$ and $x = 9 \text{ m}$, with bridge entry and exit indicated.

2. Time-Domain Response

Before 3 s the bridge acceleration is essentially zero. Once the vehicle enters the bridge, both locations begin to vibrate and the response grows as the vehicle progresses across the span. The larger oscillations occur during the later part of the crossing, particularly near 5.0-5.5 s. After 6 s, a continuing oscillatory response is still visible, indicating that the structural response does not immediately disappear when the moving load leaves the bridge.

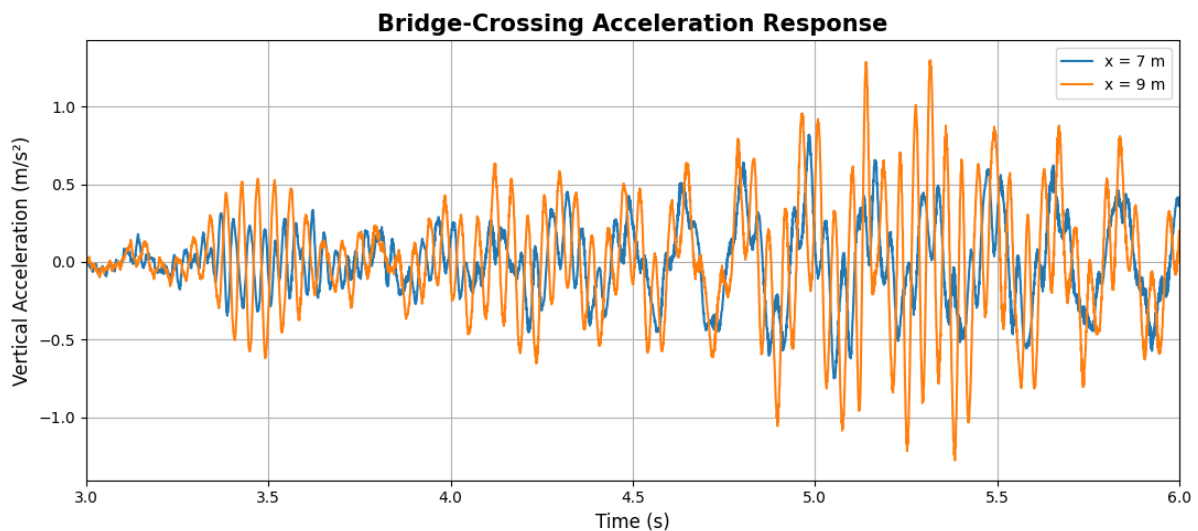


Figure 2. Bridge-crossing acceleration response during the 3-6 s vehicle passage.

During the bridge crossing, $x = 9 \text{ m}$ produces the stronger overall response. The peak absolute acceleration is 1.296592 m/s^2 at $x = 9 \text{ m}$ compared with 0.818131 m/s^2 at $x = 7 \text{ m}$, an increase of about **58.5%**. The RMS acceleration is also higher at $x = 9 \text{ m}$: 0.381683 m/s^2 compared with 0.247130 m/s^2 , about **54.4%** higher. Therefore, the 9 m position experiences a clearly stronger local vibration response during the vehicle passage.

3. Frequency-Domain Analysis

The FFT was calculated using only the bridge-crossing segment from 3 s to 6 s. The spectrum below compares the important frequency content at the two measurement locations over 0-50 Hz.

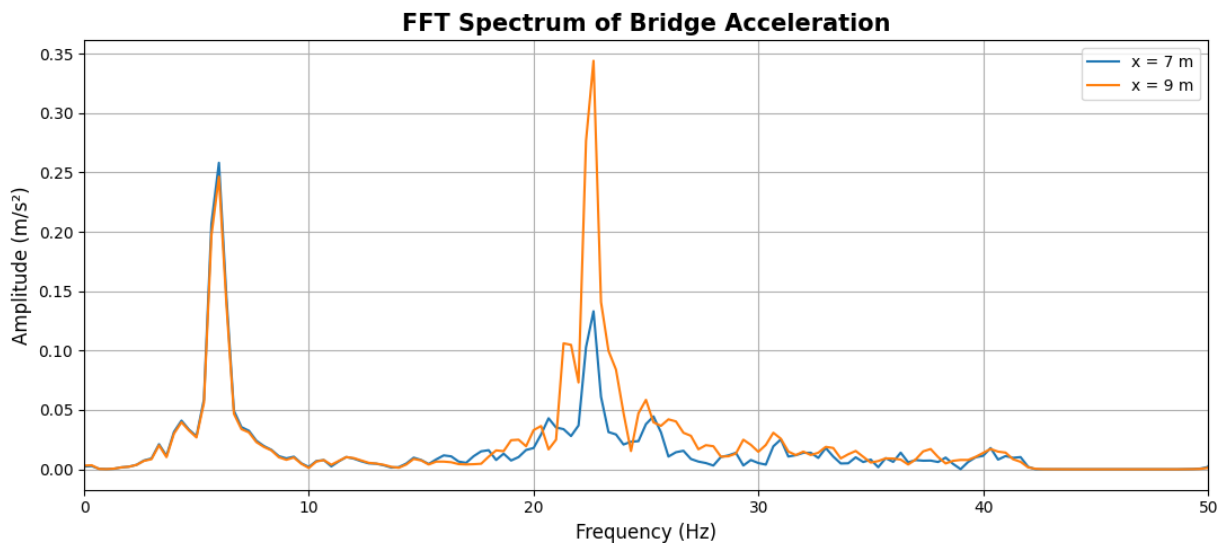


Figure 3. FFT spectrum of bridge acceleration for $x = 7$ m and $x = 9$ m.

Quantity	$x = 7$ m	$x = 9$ m
Peak absolute acceleration	0.818131 m/s ²	1.296592 m/s ²
RMS acceleration	0.247130 m/s ²	0.381683 m/s ²
Dominant FFT peak	5.998 Hz 0.258072 m/s ²	22.659 Hz 0.343943 m/s ²
Second major FFT peak	22.659 Hz 0.133054 m/s ²	5.998 Hz 0.246405 m/s ²

4. FFT Results and Engineering Interpretation

At $x = 7$ m, the largest FFT component occurs at **5.998 Hz** with an amplitude of **0.258072 m/s²**. A second strong component occurs at **22.659 Hz** with an amplitude of **0.133054 m/s²**. At $x = 9$ m, the dominant component changes to **22.659 Hz** with an amplitude of **0.343943 m/s²**, while the **5.998 Hz** component remains strong at **0.246405 m/s²**. Thus, both locations contain the same two principal frequency components, but their relative amplitudes differ substantially. The 22.659 Hz component at $x = 9$ m is about **2.58 times** the corresponding amplitude at $x = 7$ m, whereas the 5.998 Hz component is only slightly stronger at $x = 7$ m.

The remaining identified peaks are smaller: $x = 7$ m has peaks at 25.325, 20.660 and 4.332 Hz, while $x = 9$ m has peaks at 21.326, 24.992 and 25.991 Hz. The common frequency content suggests that both locations respond to the same moving vehicle and structural system, but the magnitude of each component depends on measurement position. This is consistent with spatially varying bridge mode shapes and vehicle-bridge interaction. Importantly, an FFT peak should **not** automatically be labelled as a bridge natural frequency; the response may include structural vibration, moving-load excitation, road-surface effects and vehicle-bridge interaction.

5. Conclusion

The analysis shows a clear difference in vibration response between $x = 7$ m and $x = 9$ m. The 9 m location has both the larger peak acceleration and the larger RMS acceleration, demonstrating a stronger overall bridge-crossing response. The FFT results show two major common frequency components at approximately 6 Hz and 22.659 Hz; however, the 22.659 Hz component is much stronger at $x = 9$ m. Overall, the results demonstrate that bridge vibration is strongly dependent on measurement position even under the same vehicle passage, and that both time-domain and frequency-domain measures are necessary for a complete comparison.

Key quantitative findings: Peak acceleration: 0.818131 vs 1.296592 m/s²; RMS: 0.247130 vs 0.381683 m/s²; shared major FFT peaks: 5.998 Hz and 22.659 Hz.